

Teaching Exam

Subject – Computer Science

Topic – Computer Network

Lecture No.- 10



By– Rahul Sir

Topics to be Covered

Topic



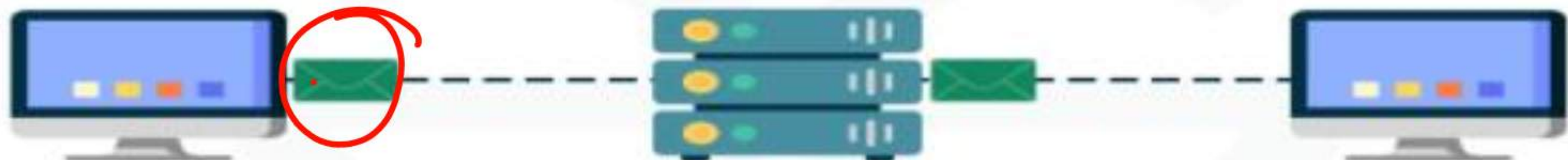
- 1:- SMTP PROTOCOL AND ITS WORKING MODEL
- 2:- POP3 PROTOCOL ✓
- 3:- IMAP PROTOCOL ✓
- 4:- Difference between POP3 and IMAP Protocol ✓
- 5:- HTTP PROTOCOL ✓
- 6:- HTTPS PROTOCOL ✓
- 7:- Difference between HTTP & HTTPS ✓

SMTP represents **Simple Mail Transfer Protocol**. SMTP is a set of interaction guidelines that allow the software to transmit electronic mail over the internet, referred to as Simple Mail Transfer Protocol.

The main objective of SMTP is used to set up communication rules between servers. The servers have a way of recognizing themselves and announcing what kind of communication they are trying to perform. They also have a way of handling errors such as incorrect email address. For example, if the recipient address is wrong, then receiving a server reply with an error message of some kind.

To send an email, the client opens a TCP connection to the SMTP server. The server, which is always listening on port 25, initiates the connection as soon as it detects a client. Once the TCP connection is established, the client sends the email across the connection.

SMTP

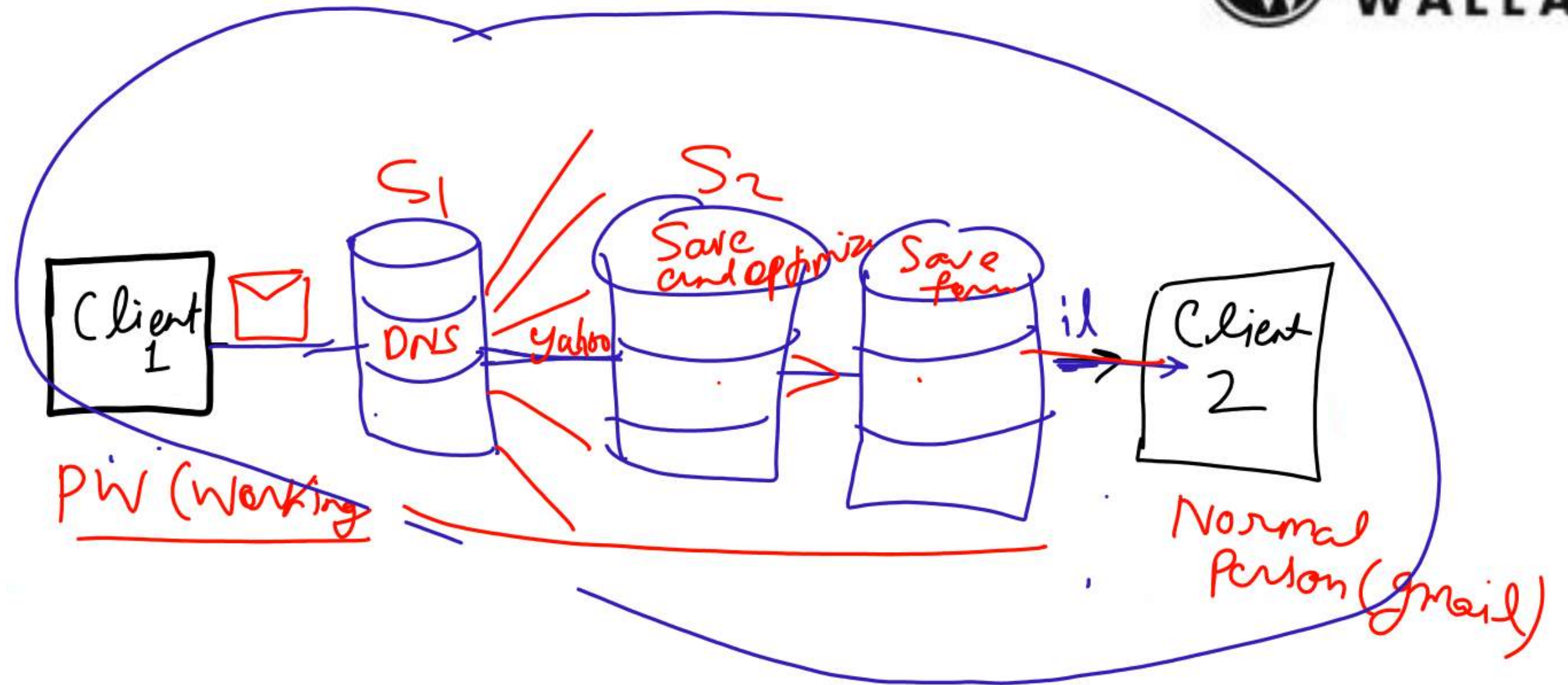


Types of SMTP Protocol

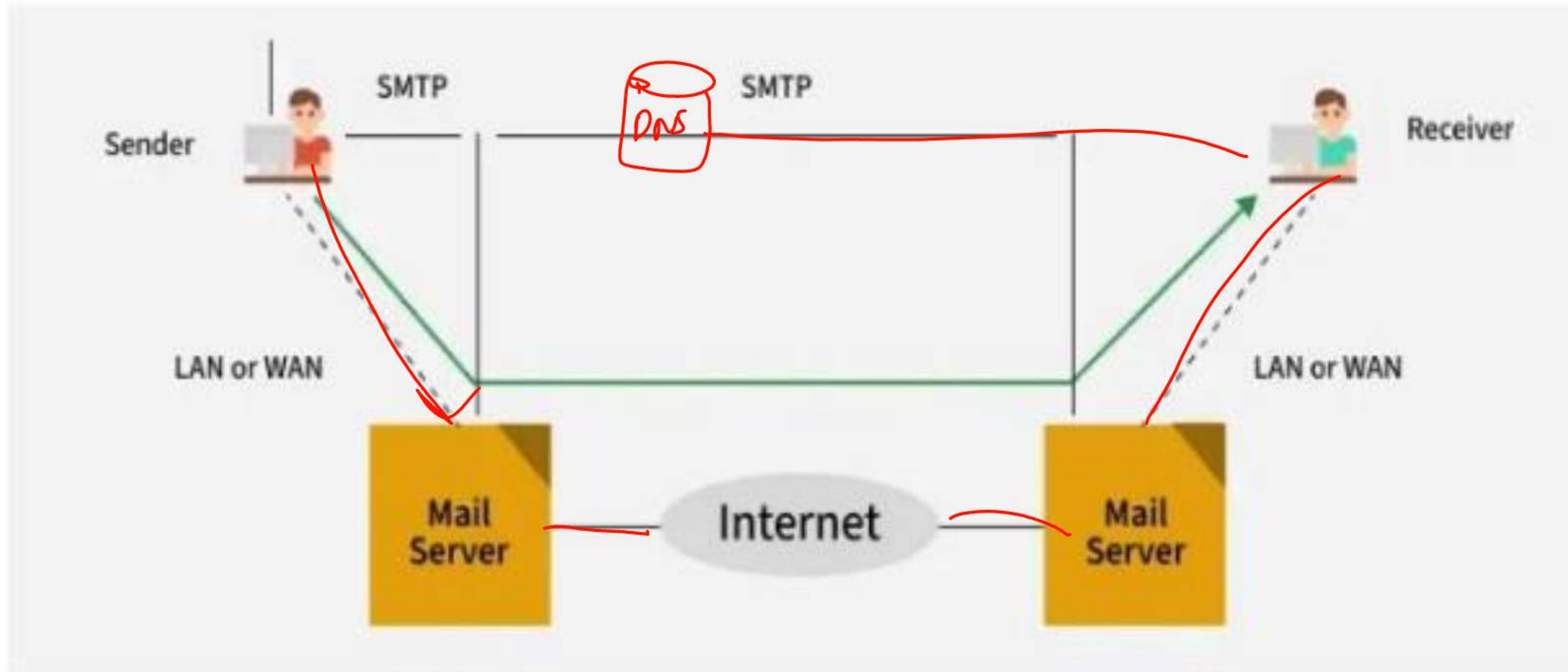
The SMTP model supports two types of email delivery methods: **end-to-end** and **store-and-forward**.

- **End-to-end** delivery is used between organizations. In this method, the email is sent directly from the sender's SMTP client to the recipient's SMTP server without passing through intermediate servers.
- **Store-and-forward** is used within organizations that have TCP/IP and SMTP-based networks. In this method, the email may pass through several intermediate servers (Message Transfer Agents, or MTAs) before reaching the recipient.

With **end-to-end** delivery, the SMTP client waits until the email is successfully copied to the recipient's SMTP server before sending it. This is different from the **store-and-forward** method, where the email might stop at multiple intermediate servers before reaching its destination. In store-and-forward systems, the sender is notified as soon as the email reaches the first server, not the final destination.

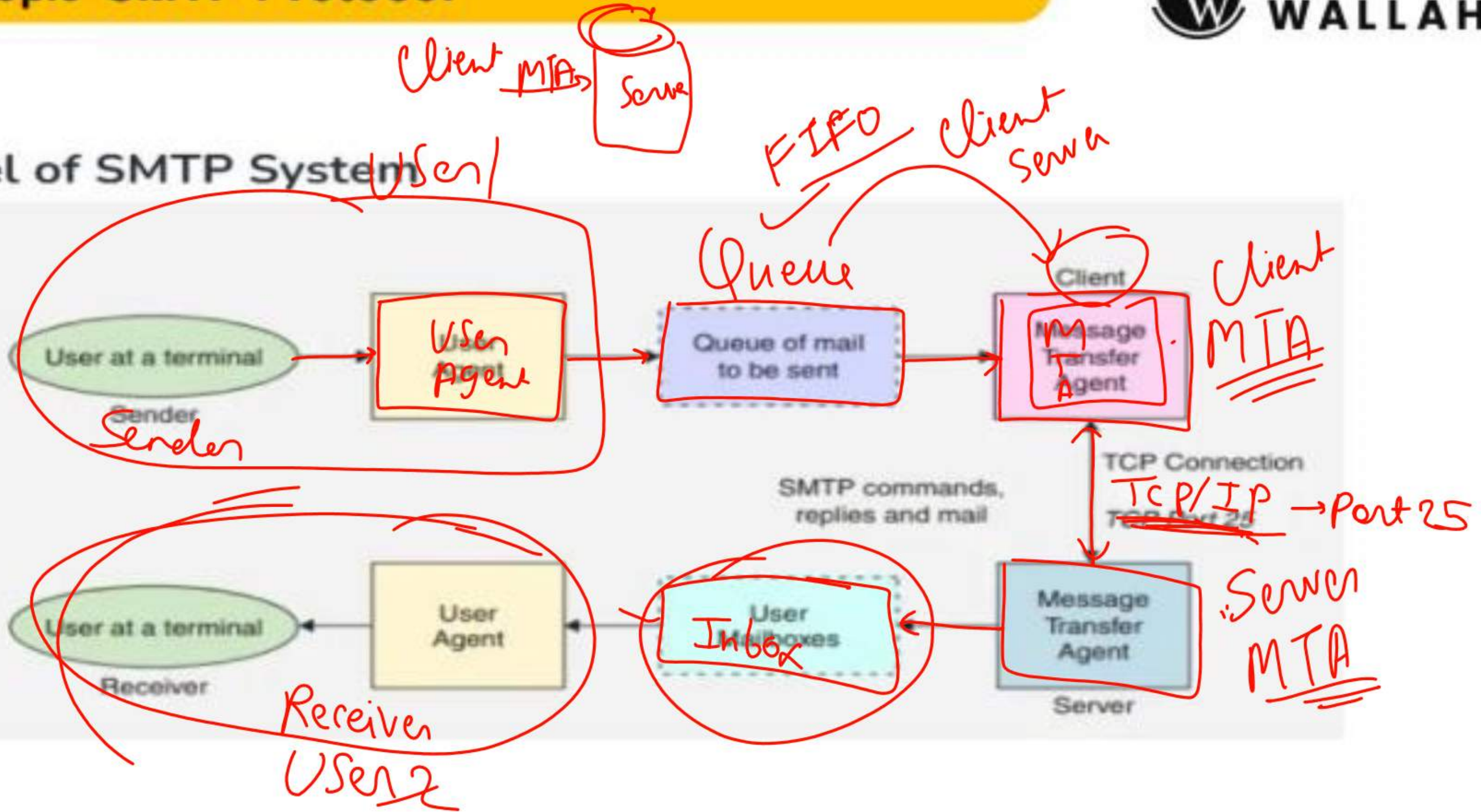


Topic- SMTP Protocol



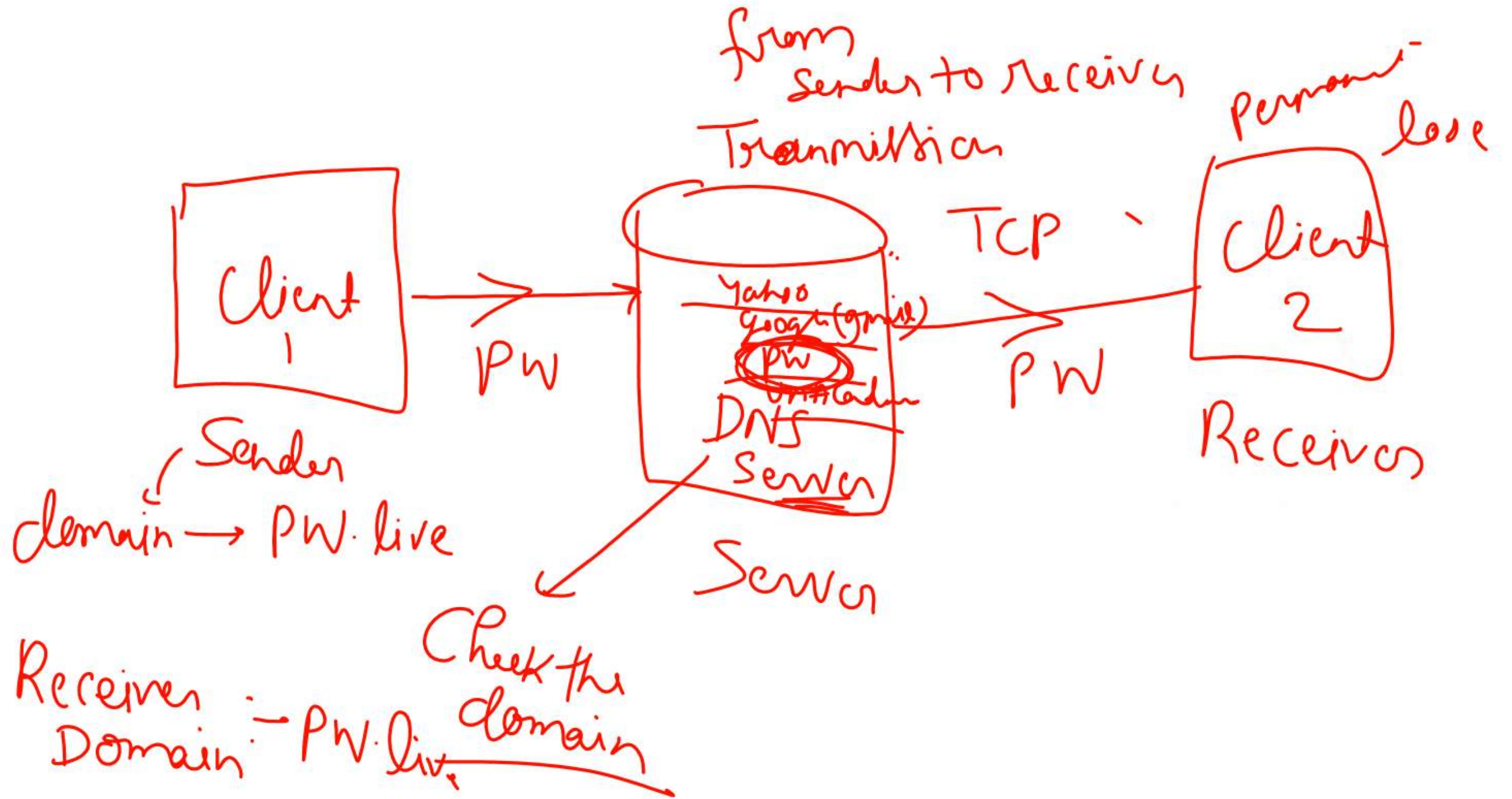
Topic- SMTP Protocol

Model of SMTP System



Introduction to POP

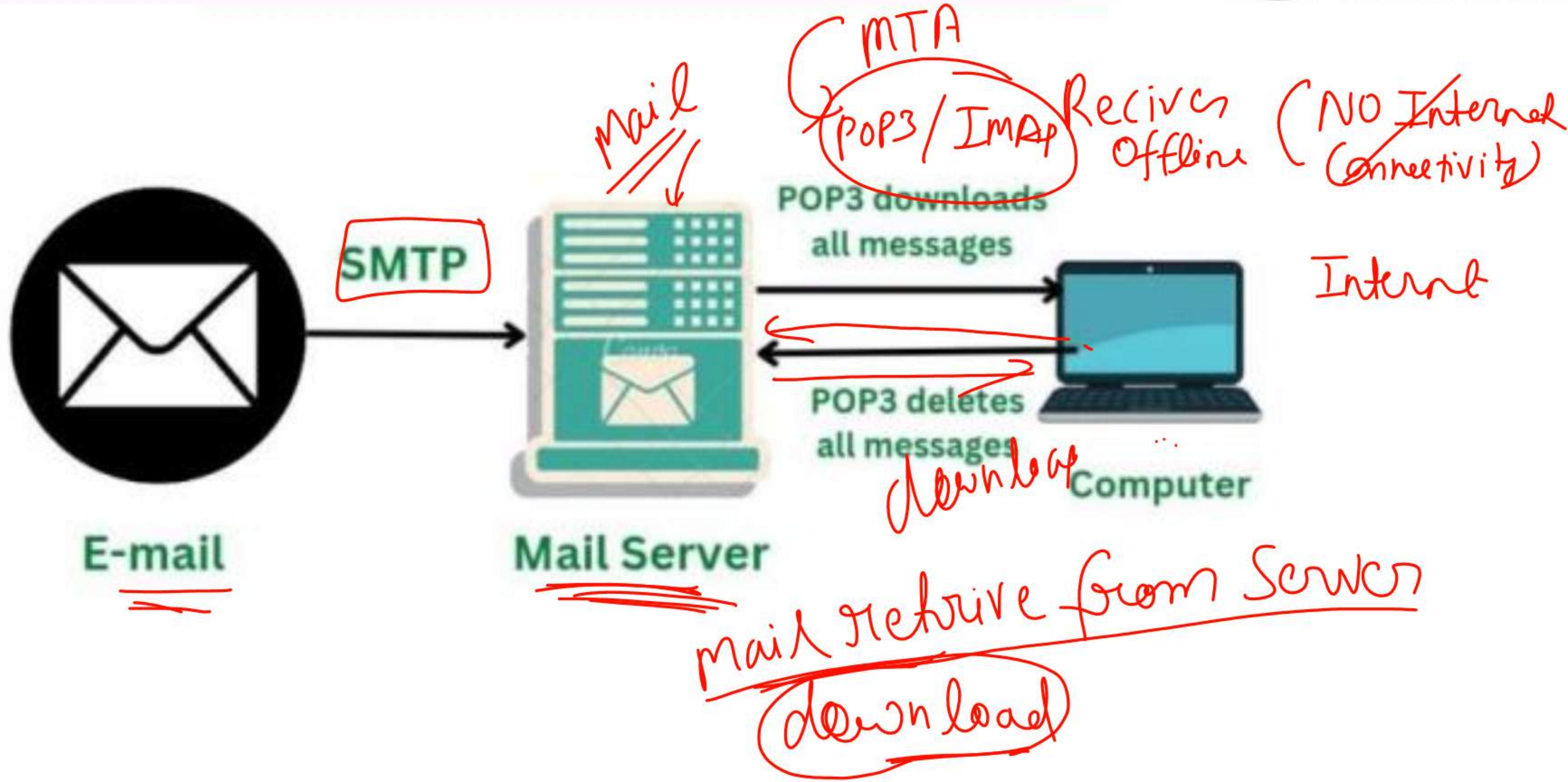
POP stands for Post Office Protocol. The [POP protocol](#) was published in the year 1984. POP has been updated two times namely "POP2" and "POP3". The POP protocol is an Internet Standard Protocol that works on the application layer. It is used to get an access email from the mail server. The need for POP mainly arises when the user or client does not have a continuous internet connection and wants to receive email messages. The Pop client makes use of POP to pull email messages from the POP server. POP3 is the updated version of POP.



What is POP3?

POP 3 stands for Post Office Protocol Version 3. POP3 protocol is used to provide access to the mail inbox that is stored in the email server. POP3 protocol can download and delete messages. Once the POP3 client has established a connection with the mail server it can easily retrieve all the messages from the server. The user can access the messages locally even if the user is offline. The protocol does not inherently support real-time synchronization or automatic checking for new messages, users can configure their email clients to check for new messages at intervals or manually. Many email programs, including Apple Mail, Gmail, and Microsoft Outlook, support the POP3 protocol, although IMAP is often preferred for its synchronization features.

When a message is sent, SMTP is used to transfer it from the client to the server and ultimately to the server of the recipient. However, the Message Access Agent facilitates the transmission of the message from the receiving server to the host server. POP3 and IMAP are the two types of protocols that are included in the Message Access Agent.



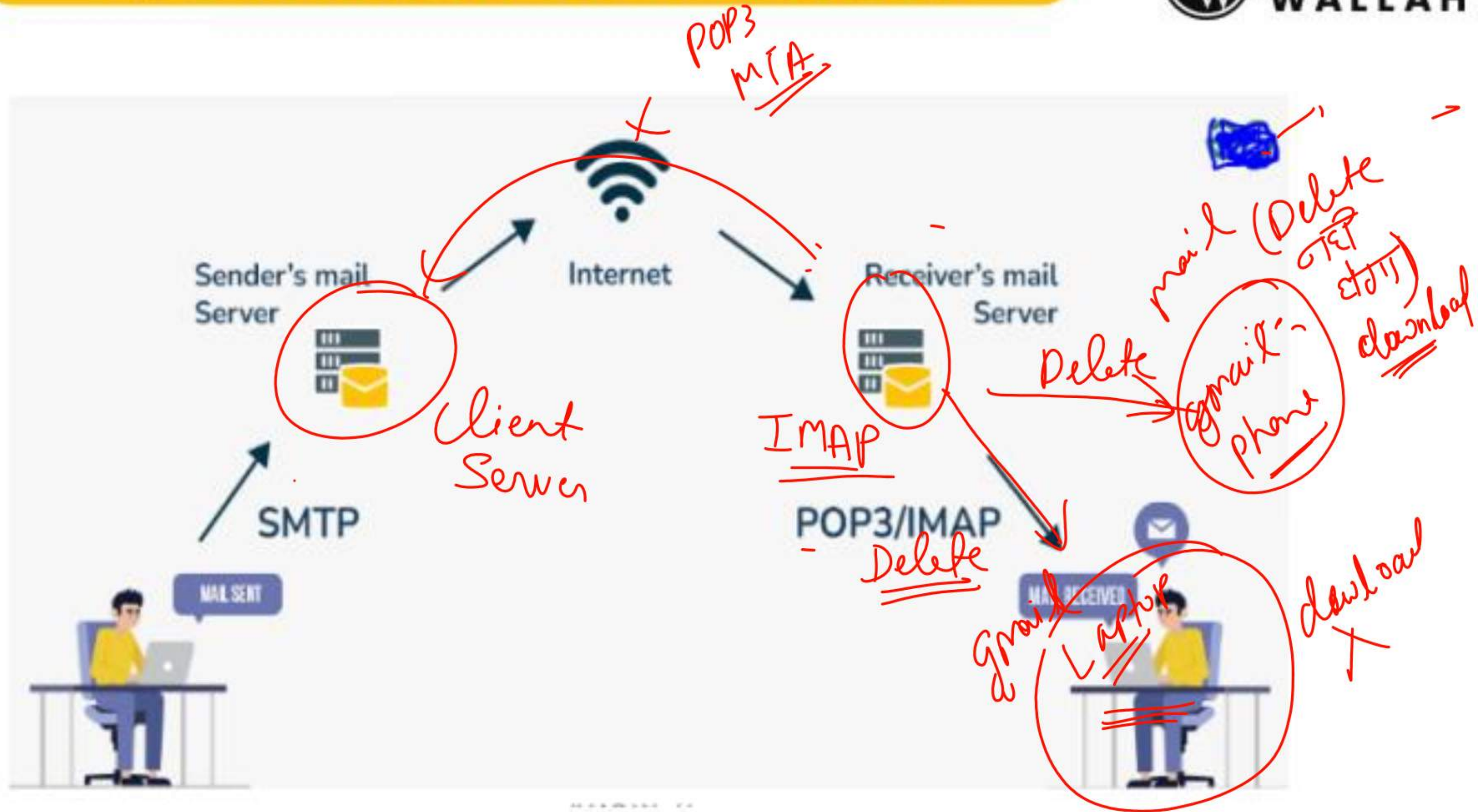
Advantages of POP3

- The message can be read offline also.
- POP3 requires less storage space.
- POP3 is easy to use and configure.
- POP3 is supported by many email applications.
- Since our PC already has the emails stored there, accessing them is quick and simple.
- The size of emails we send and receive is unrestricted.
- Since all emails are kept locally, less server storage space is needed.

Disadvantages of POP3

- POP3 does not provide the feature of real-time synchronization.
- If an email consists of virus attachments it can affect the system easily.
- POP3 does not support accessing the same email at the same time on different systems.
- At a time entire email folder can be discarded.
- ✖ It is also possible for the email folder that is downloaded from the mail server to get

Topic- IMAP Protocol – EMAIL PROTOCOL



Internet Message Access Protocol (IMAP) is an application layer protocol that operates as a contract for receiving emails from the mail server. It was designed by Mark Crispin in 1986 as a remote access mailbox protocol, the current version of IMAP is IMAP4.

It is used as the most commonly used protocol for retrieving emails. This term is also known as Internet mail access protocol, Interactive mail access protocol, and Interim mail access protocol.

It retrieves messages from your email provider to your email client (such as on your Mac or PC). Importantly, it does not delete messages from the email service after you download them until the user explicitly deletes them.

It allows you to view and manage your emails across multiple devices because it synchronizes the emails between your email client and the server. This synchronization ensures that changes such as read status, deletions, and folder organization are reflected consistently across all devices connected to the same email account.

Working of IMAP

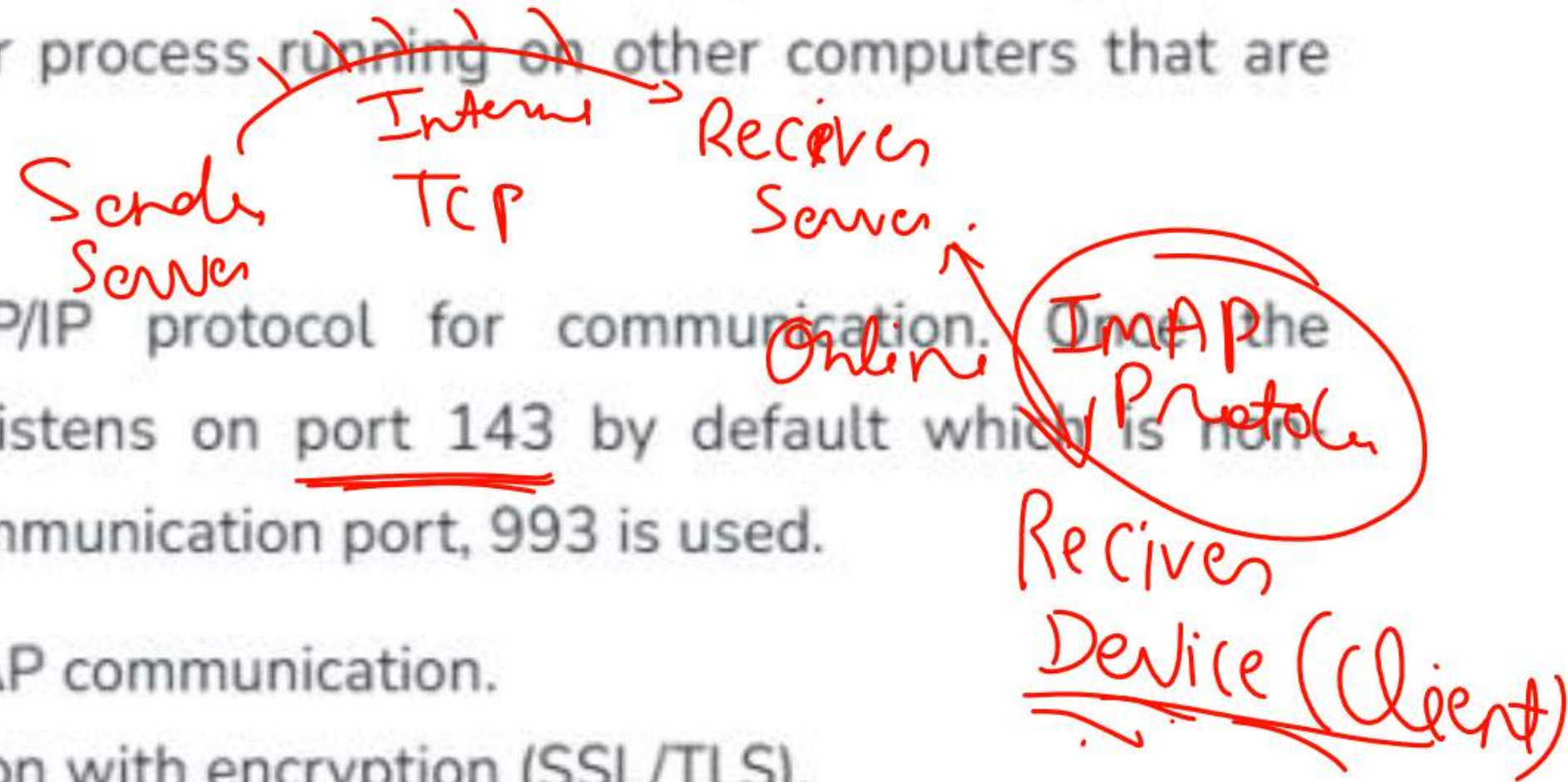
IMAP follows Client-server Architecture and is the most commonly used email protocol. It is a combination of client and server process running on other computers that are connected through a network.

IMAP: → 143 Port

This protocol resides over the TCP/IP protocol for communication. Once the communication is set up the server listens on port 143 by default which is non-encrypted. For the secure encrypted communication port, 993 is used.

TCP: → 25 Port

- **Port 143:** Default non encrypted IMAP communication.
- **Port 993:** Secure IMAP communication with encryption (SSL/TLS).



Advantages

- It offers synchronization across all the maintained sessions by the user.
- It provides security over POP3 protocol as the email only exists on the IMAP server.
- Users have remote access to all the contents.
- It offers easy migration between the devices as it is synchronized by a centralized server.
- There is no need to physically allocate any storage to save contents.

Disadvantages

- IMAP is complex to maintain. ✓
- Emails of the user are only available when there is an internet connection.
- It is slower to load messages. ✓
- Some emails don't support IMAP which makes it difficult to manage. ✓

Topic- Difference between POP3 and IMAP Protocol

Post Office Protocol (POP3)

POP → 110 Port no.

IMAP → 143 Port no.

POP is a simple protocol that only allows downloading messages from your Inbox to your local computer.

TCP/IP → 25 Port

The POP server listens on port 110, and the POP with SSL secure(POP3DS) server listens on port 995

In POP3 the mail can only be accessed from a single device at a time.

To read the mail it has to be downloaded on the local system.

Internet Message Access Protocol (IMAP)

IMAP (Internet Message Access Protocol) is much more advanced and allows the user to see all the folders on the mail server.

The IMAP server listens on port 143, and the IMAP with SSL secure(IMAPDS) server listens on port 993.

Messages can be accessed across multiple devices

The mail content can be read partially before downloading.

loading

Topic- Difference between POP3 and IMAP Protocol

In POP3 the mail can only be accessed from a single device at a time.	Messages can be accessed across multiple devices
To read the mail it has to be downloaded on the local system.	The mail content can be read partially before downloading.
The user can not organize mail in the mailbox of the mail server.	On the mail server, the user can directly arrange the email.
The user can not create, delete, or rename email on the mail server.	The user can create, delete, or rename an email on the mail server.
It is unidirectional i.e. all the changes made on a device do not affect the content present on the server.	It is Bi-directional i.e. all the changes made on the server or device are made on the other side too.
It does not allow a user to sync emails.	It allows a user to sync their emails.

Changes in the mail can be done using local email software.	Changes made to the web interface or email software stay in sync with the server.
All the messages are downloaded at once.	The Message header can be viewed before downloading.

HTTP (Hypertext Transfer Protocol) is a fundamental protocol of the Internet, enabling the transfer of data between a client and a server. It is the foundation of data communication for the World Wide Web.

HTTP provides a standard between a web browser and a web server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.



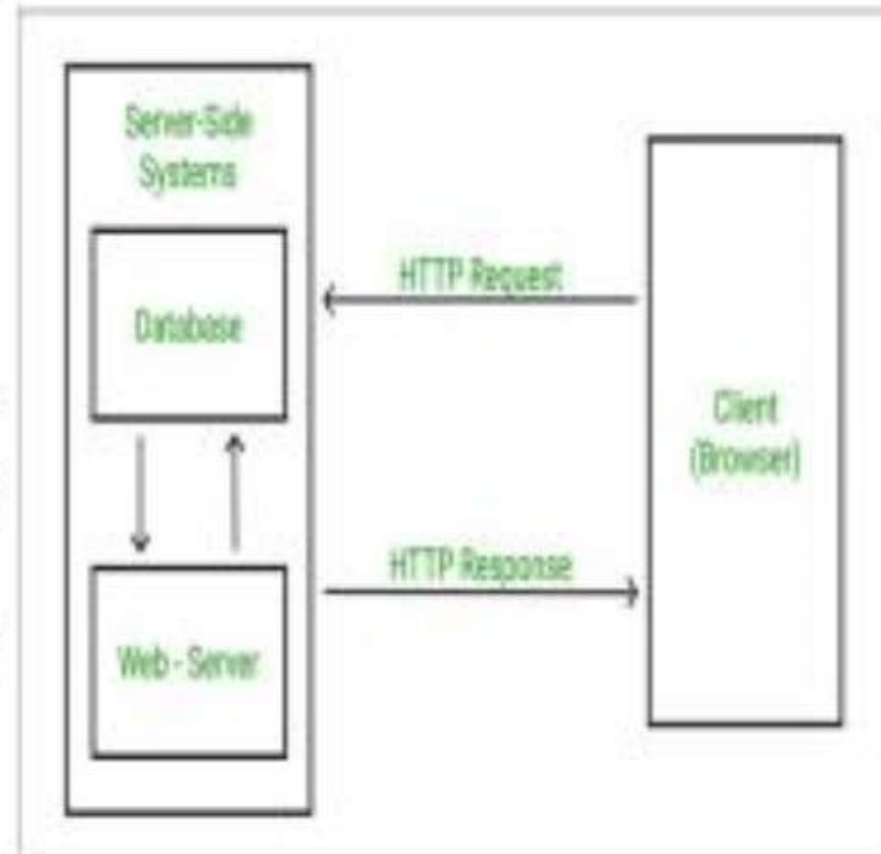
Methods of HTTP

- **GET:** Used to retrieve data from a specified resource. It should have no side effects and is commonly used for fetching web pages, images, etc.
- **POST:** Used to submit data to be processed by a specified resource. It is suitable for form submissions, file uploads, and creating new resources.
- **PUT:** Used to update or create a resource on the server. It replaces the entire resource with the data provided in the request body.
- **PATCH:** Similar to PUT but used for partial modifications to a resource. It updates specific fields of a resource rather than replacing the entire resource.
- **DELETE:** Used to remove a specified resource from the server.

HTTP Request/Response:

HTTP is a request-response protocol, which means that for every request sent by a client (typically a web browser), the server responds with a corresponding response. The basic flow of an HTTP request-response cycle is as follows:

- **Client sends an HTTP request:** The client (usually a web browser) initiates the process by sending an HTTP request to the server. This request includes a request method (GET, POST, PUT, DELETE, etc.), the target URI (Uniform Resource Identifier, e.g., a URL), headers, and an optional request body.
- **Server processes the request:** The server receives the request and processes it based on the requested method and resource. This may involve retrieving data from a database, executing server-side scripts, or performing other operations.
- **Server sends an HTTP response:** After processing the request, the server sends an HTTP response back to the client. The response includes a status code (e.g., 200 OK, 404 Not Found), response headers, and an optional response body containing the requested data or content.
- **Client processes the response:** The client receives the server's response and processes it accordingly. For example, if the response contains an HTML page, the browser will render and display it. If it's an image or other media file, the browser will display or handle it appropriately.



Concept	Description
HTTP Request Headers	Text information in key-value pairs sent with every HTTP request, conveying details such as client's browser, requested data, and more.
HTTP Request Body	Contains data being submitted to the web server, such as form inputs, authentication details, or other payload data.
HTTP Response	Sent by the server in answer to an HTTP request, consisting of an HTTP status code, response headers, and optional response body.
HTTP Status Codes	3-digit codes indicating the outcome of an HTTP request, categorized into 1xx (Informational), 2xx (Success), 3xx (Redirection), 4xx (Client Error), and 5xx (Server Error) blocks.
HTTP Response Headers	Provide important information like data format, language, and content type in the response, helping clients interpret the data correctly.
HTTP Response Body	Contains the requested information sent by the server, such as HTML content for webpages, JSON data, or other data based on the client's request.

Advantages

- **Platform independence:** Works on any operating system
- **Compatibility:** Compatible with various protocols and technologies
- **Efficiency:** Optimized for performance
- **Security:** Supports encryption for secure data transfer

Disadvantages

- **Lack of security:** Vulnerable to attacks like man in the middle
- **Performance issues:** Can be slow for large data transfers
- **Statelessness:** Requires additional mechanisms for maintaining state

HTTPS stands for HyperText Transfer Protocol Secure. It is the most common protocol for sending data between a web browser and a website. HTTPS is the secure variant of HTTP and is used to communicate between the user's browser and the website, ensuring that data transfer is encrypted for added security.

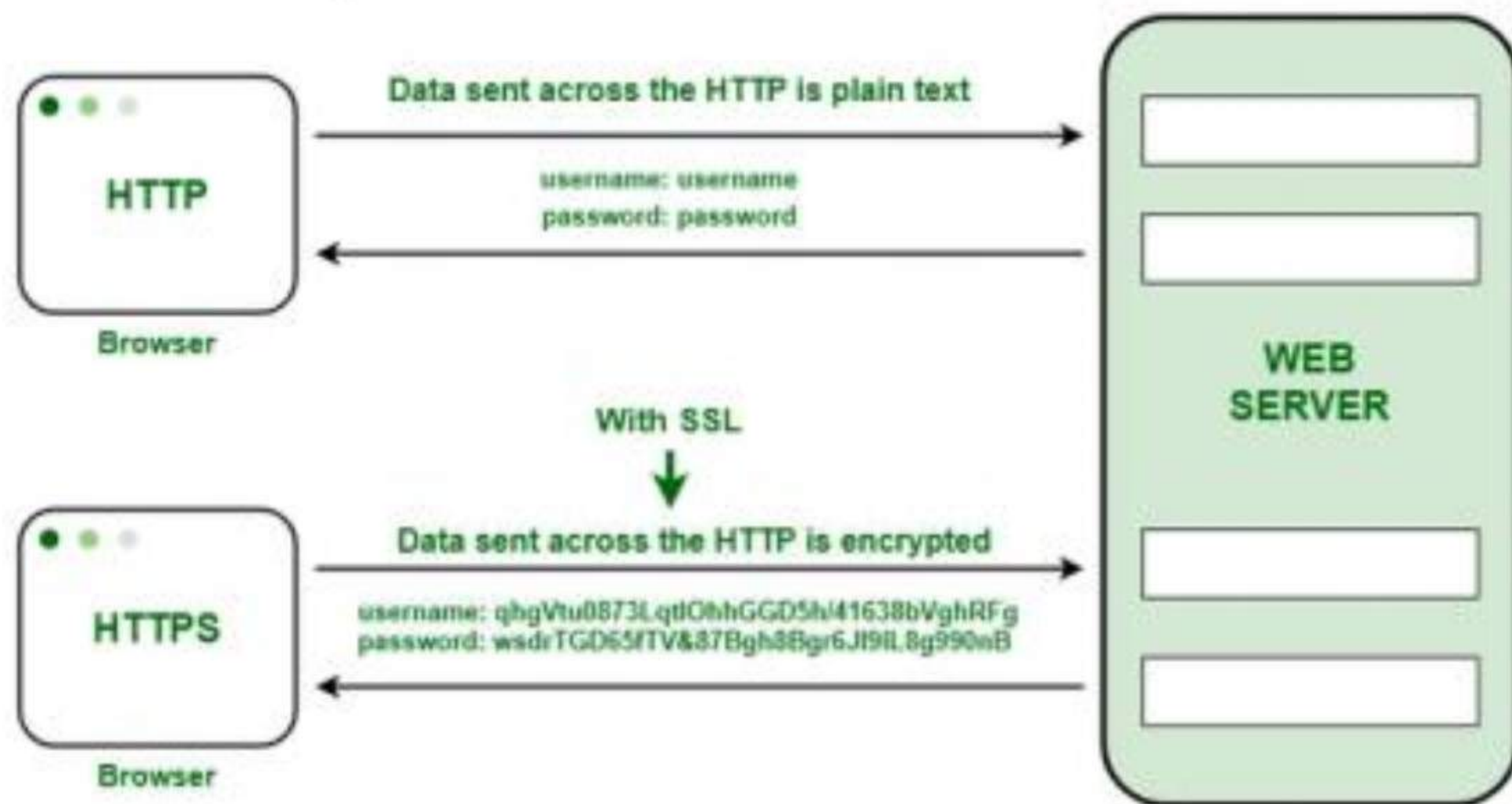
Any website, especially those requiring login details, should use HTTPS. You can see a padlock icon in the URL bar, which means the page is secure. Browsers, like Google Chrome, treat HTTPS seriously and mark non-HTTPS websites as "Not Secure."

HTTPS



HTTPS establishes the communication between the browser and the web server. It uses the **Secure Socket Layer (SSL)** and **Transport Layer Security (TLS)** protocol for establishing communication. The new version of SSL is TLS(Transport Layer Security).

HTTPS uses the conventional HTTP protocol and adds a layer of SSL/TLS over it. The [workflow of HTTP and HTTPS](#) remains the same, the browsers and servers still communicate with each other using the HTTP protocol. However, this is done over a secure SSL connection. The SSL connection is responsible for the encryption and decryption of the data that is being exchanged to ensure data safety.



Why HTTPS Matters and What Happens Without It?

HTTPS is important because it keeps the information on websites safe from being easily viewed or stolen by anyone who might be spying on the network. When a website uses regular HTTP, data is sent in small chunks called packets that can easily be intercepted using free software. This makes communication, especially over public Wi-Fi, very vulnerable to attacks.

On the other hand, HTTPS encrypts the data, so even if someone manages to intercept the packets, they will appear as random, unreadable characters. For example:

- **Before encryption:**

```
"This is a string of text that is completely readable"
```

- **After encryption:**

```
"ITM0IRyiEhVpa6VnKyExMiEgNveroyWBP1gGyfkflYjDaaFf/Kn3bo3OfghBPDWo6AfSH1NtL8N7ITEwIXc1gU5X73xMsJormzzXlwOyrC  
s+9XCPk63Y+z0="
```

Secure Socket Layer (SSL)

The main responsibility of SSL is to ensure that the data transfer between the communicating systems is **secure and reliable**. It is the standard security technology that is used for encryption and decryption of data during the transmission of requests.

As discussed earlier, HTTPS is basically the same old HTTP but with SSL. For establishing a secure communication link between the communicating devices, SSL uses a digital certificate called **SSL certificate**.

There are two major roles of the SSL layer

- Ensuring that the browser communicates with the required server directly.
- Ensuring that only the communicating systems have access to the messages they exchange.

Advantage of HTTPS

- **Secure Communication:** HTTPS establishes a secure communication link between the communicating system by providing encryption during transmission.
- **Data Integrity:** By encrypting the data, HTTPS ensures data integrity. This implies that even if the data is compromised at any point, the hackers won't be able to read or modify the data being exchanged.
- **Privacy and Security:** HTTPS prevents attackers from accessing the data being exchanged passively, thereby protecting the privacy and security of the users.
- **Faster Performance:** TTPS encrypts the data and reduces its size. Smaller size accounts for faster data transmission in the case of HTTPS.

HTTP vs HTTPS

Below are the basic differences between HTTP and HTTPS.

HTTP	HTTPS
HTTP stands for HyperText Transfer Protocol.	HTTPS stands for HyperText Transfer Protocol Secure.
URL begins with "http://".	URL starts with "https://".
HTTP Works at the Application Layer .	HTTPS works at Transport Layer .
HTTP speed is faster than HTTPS.	HTTPS speed is slower than HTTP.

THANK YOU